Paleomagnetic and geobarometric study of the mid-Cretaceous Whitehorse Pluton, Yukon Territory[1]

M.J. Harris, D.T.A. Symons, W.H. Blackburn, and C.J.R. Hart

Abstract: This is the first of several Lithoprobe paleomagnetic studies underway to examine geotectonic motions in the northern Canadian Cordillera. Except for one controversial study, estimates for terranes underlying the Intermontane Belt in the Yukon have been extrapolated from studies in Alaska, southern British Columbia, and the northwestern United States. The Whitehorse Pluton is a large unmetamorphosed and undeformed tonalitic body of mid-Cretaceous age (~112 Ma) that was intruded into sedimentary units of the Whitehorse Trough in the Stikinia terrane. Geothermobarometric estimates for eight sites around the pluton indicate that postmagnetization tilting has been negligible since cooling through the hornblende-crystallization temperature and that the pluton is a high-level intrusion. Paleomagnetic measurements for 22 of 24 sites in the pluton yield a well-defined characteristic remanent magnetization (ChRM) direction that is steeply down and northwards. The ChRM direction gives a paleopole of 285.5°E, 81.7°N (d_p = 5.3°, d_m = 5.7°). When compared with the 112 Ma reference pole for the North American craton, this paleopole suggests that the northern Stikinia terrane has been translated northwards by 11.0 ± 4.8° (1220 ± 530 km) and rotated clockwise by 59 ± 17°. Except for an estimate from the ~70 Ma Carmacks Group volcanics, this translation and rotation estimate agrees well with previous estimates for units in the central and southern Intermontane Belt. They suggest that the terranes of the Intermontane Belt have behaved as a fairly coherent unit since the Early Cretaceous, moving northward at a minimum average rate of 2.3 ± 0.4 cm/a between ~140 and ~45 Ma.

Résumé : Nous présentons ici la première de plusieurs études paléomagnétiques du projet Lithoprobe entreprises pour estimer les mouvements géotectoniques dans la Cordillère septentrionale au Canada. À l'exception d'une étude controversée, les estimations connues pour les terranes sous-jacents au Domaine intermontagneux dans le Yukon sont des extrapolations dérivées d'études faites dans les régions de l'Alaska, du sud de la Colombie-Britannique et du nord-ouest des États-Unis. Le pluton de Whitehorse est un puissant corps tonalitique postmétamorphique et post-tectonique, daté du Crétacé moyen (~112 Ma), qui a fait intrusion dans les unités sédimentaires de la cuvette de Whitehorse dans le terrane de Stikinia. Les estimations géothermobarométriques de huit sites autour du pluton l'absence de gauchissement important postérieur à l'acquisition de l'aimantation, depuis la phase de refroidissement sous le seuil de la température de cristallisation de la hornblende, et en plus elles indiquent une mise en place du pluton à un niveau élevé. Les mesures paléomagnétiques de 22 sur 24 sites dans le pluton fournissent une direction caractéristique de rémanence bien définie plongeant abruptement et vers le nord. La direction caractéristique de rémanence est caractérisée par un paléopôle de 285,5°E, 81,7°N (d_p = 5,3°, d_m = 5,7°). Ce paléopôle, lorsque comparé au pôle de référence de 112 Ma pour le craton de l'Amérique du Nord, suggère que la partie septentrionale du terrane de Stikinia a glissé par translation vers le nord de 11,0 ± 4,8° (1220 ± 530 km) avec une rotation horaire de 59 ± 17°. À part une estimation pour les volcanites du Groupe de Carmacks de ~70 Ma, les valeurs estimées de la translation et de la rotation sont en accord avec celles proposées précédemment pour les unités des régions centrale et méridionale du Domaine intermontagneux. Elles suggèrent que les terranes du Domaine intermontagneux se sont comportés comme une unité passablement cohérente depuis le Crétacé précoce, se déplaçant vers le nord avec une vitesse moyenne minimum de 2,3 ± 0,4 cm/a entre ~140 et ~45 Ma.
[Traduit par la rédaction]

Introduction

The ~112 Ma Whitehorse Pluton is a tonalitic body that crops out in the northern Intermontane Belt to the southwest of Whitehorse, Yukon Territory (Fig. 1). It is one of several targets for Lithoprobe paleomagnetic studies being done at the University of Windsor to better understand the geotectonic history of the northern Cordillera.

The Carmacks Group volcanics is the only mid-Mesozoic to Eocene unit in the Yukon that has been studied paleomagnetically and indicates that the Whitehorse Trough has been displaced northward by 1900 ± 700 km between 70 and 50 Ma (Johnston et al. 1996). This estimate is inconsistent with most values extrapolated from the Intermontane Belt to the south. Here we report paleomagnetic data from the Whitehorse Pluton as well as geothermobarometric data on hornblende and plagioclase crystals to test for postmagnetization tilting of the pluton, in order to estimate the northward displacement of the Whitehorse Trough relative to the North American craton.

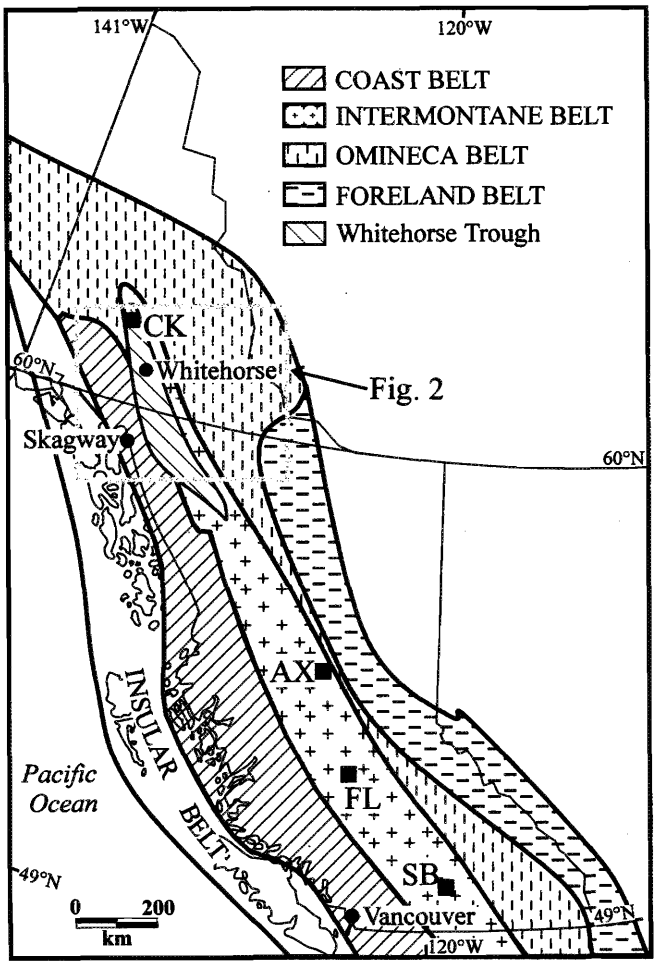
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Fig. 1. The morphogeological belts of the Canadian Cordillera. CK, Carmacks Group volcanics; AX, Axelgold intrusions; FL, Francois Lake (Topley) intrusions; SB, Spences Bridge volcanics.



Geology

Regional

The Intermontane Belt is one of five morphogeological belts in the Cordillera that are defined by a combination of lithological, structural, tectonic, and physiographic attributes (Fig. 1) (Gabrielse et al. 1991*a*). It is largely coextensive with the Stikinia, Quesnellia, and Cache Creek terranes (Fig. 2). These terranes consist of upper Paleozoic to lower Mesozoic marine assemblages, including oceanic, island-arc, foredeep, and clastic-wedge sequences, which have been intruded by a wide range of plutonic rocks since the Late Triassic (Gabrielse et al. 1991*b*; Woodsworth et al. 1991). The Whitehorse Pluton intrudes the dominantly sedimentary Whitehorse Trough in Stikinia (Fig. 1).

Stikinia is comprised of Lower Devonian to Middle Jurassic sedimentary and volcanic strata that were intruded by cogenetic plutonic rocks (Monger et al. 1991). It is in fault contact on the east with the Cache Creek Terrane, but the original nature of its western contact has been masked by Cretaceous and Tertiary plutonism and metamorphism of the Coast Plutonic Complex. The upper Paleozoic and Triassic rocks of Stikinia are thought to reflect an island-arc setting, with its 400 km long belt of Mississippian and Permian limestones recording an interval when the terrane was a stable block (Gabrielse et al. 1991*b*). During the Middle Jurassic, the Cache Creek Terrane was thrust over the northern and eastern margins of Stikinia, depositing diagnostic fossiliferous limestone and chert clasts from the Cache Creek Terrane into Middle to Late Jurassic conglomerates of the Bowser Basin. Similarly, the Early Jurassic Inklin Formation of the Cache Creek Terrane contains detritus believed to be derived from the Quesnellia terrane (Monger et al. 1991). Thus, by Middle Jurassic, the Stikinia, Quesnellia, and Cache Creek terranes were at least loosely amalgamated and located next to the craton.

Previous paleomagnetic studies suggest that the terranes underlying the Insular Belt (Fig. 2) may have collided with those of the Intermontane Belt by mid-Cretaceous to Late Cretaceous time to create a fairly coherent assemblage located some 1000–2000 km south of its present position (Irving and Wynne 1991; Irving et al. 1996), with the Coast Plutonic Complex marking the intervening suture (Monger et al. 1982). Subsequently, this assemblage of terranes moved north-northwesterly in panels on dominantly dextral faults beside the North American craton, reaching their present position in the Late Eocene (~45 Ma) (Irving et al. 1996). Paleomagnetic data for Eocene (52–44 Ma) rocks throughout the Canadian Cordillera, except for the Queen Charlotte Islands (Irving and Wynne 1991), show that the Cordilleran terranes have not moved significantly with respect to the North American craton since then.

Local

The ovate Whitehorse Pluton is exposed over 160 km^2 (Fig. 3). It intrudes two members of the Late Triassic Lewes River Group. The Hancock Member comprises up to 600 m of limestones and minor siltstones, and the overlying Mandanna Member comprises up to several hundred metres of crystal-rich, volcanogenic lithic greywacke and shale with minor red, bioturbated siltstone and pebble conglomerate (Hart and Radloff 1990).

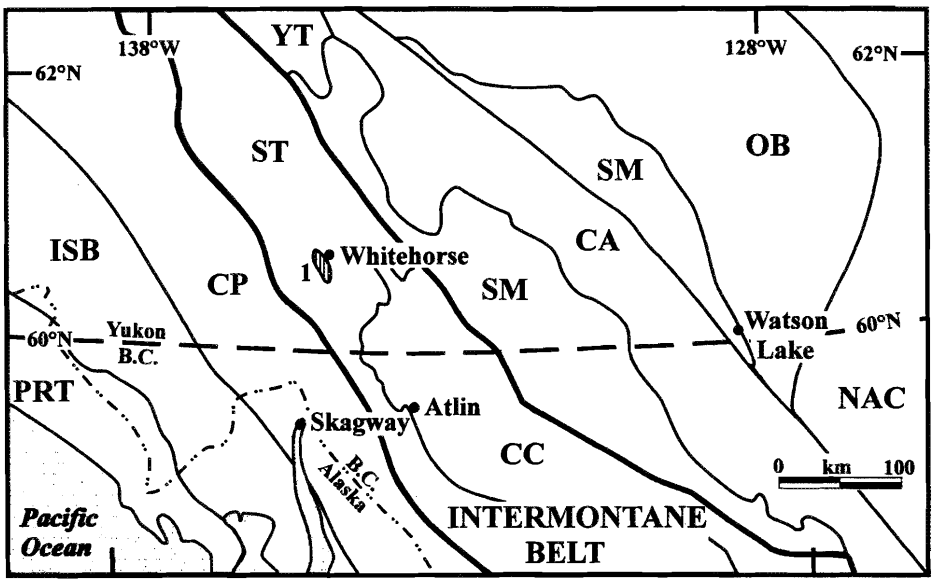
The southern part of the Whitehorse Pluton is intruded by a few narrow vertical diabase dikes of uncertain age. It is also unconformably overlain in several areas by the 8.8 Ma Miles Canyon basalt flows and then by Quaternary glaciofluvial and glaciolacustrine alluvium that obscures its eastern margin (Hart and Radloff 1990). Aeromagnetic surveys indicate its eastern contact dips steeply, and drilling indicates a steep to locally overhanging contact on the west and an outward shallow-dipping contact on its north and south sides (Morrison 1981; Hart and Pelletier 1989). Contact attitudes indicate that the pluton has not been tilted, assuming vertical emplacement. Pendants of host rock are numerous along the northern, western, and southern margins, but are not observed along the eastern margin because of lower erosional levels and the presence of the overburden.

Petrography and geochronology

The Whitehorse Pluton has been described as a synorogenic, calc-alkalic pluton (Morrison 1981) and as a postorogenic, zoned, intermediate pluton of gabbroic to granodioritic composition (Hart 1995). The interior of the pluton is generally massive, medium to light grey, and leucocratic, whereas the margins of the pluton are typically more mafic with a greater

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Fig. 2. Tectonic subdivisions of the northern Cordillera (after Wheeler et al. 1991). NAC, North American craton; OB, Omineca fold belt; CA, Cassiar Terrane; YT, Yukon–Tanana Terrane; SM, Slide Mountain Terrane; CP, Coast Plutonic Complex; ISB, Insular Belt; PRT, Pacific Rim terranes; 1, Whitehorse Pluton. INTERMONTANE BELT includes Cache Creek (CC) and Stikinia (ST) terranes (all bounded by thick lines).



density of xenoliths. Based on thin-section analyses and field observations for all paleomagnetic sites and elsewhere, the rock types include diorite, quartz diorite, quartz monzodiorite, tonalite, and granodiorite. Most have medium- to coarse-grained plagioclase in a fine- to medium-grained matrix of quartz and mafic minerals (Hart and Radloff 1990). The major minerals are plagioclase, quartz, hornblende, and biotite. Minor and accessory minerals include magnetite, potassium feldspar, titanite, and apatite. Larger hornblende crystals typically have clinopyroxene cores. The anorthite content of plagioclase ranges from An_{19} to An_{34} (average An_{25}), and some albite may be present. Visible potassium feldspar crystals are rare, but perthite is common. The pluton is variably altered, as the presence of chlorite, epidote, and carbonate suggests. Zones of hydrothermal alteration in the northern part of the pluton are probably related to adjacent copper–gold skarn deposits in that region (Morrison 1981; Hart and Pelletier 1989; Hart and Radloff 1990; Hart 1995).

Two sets of K–Ar ages on biotite and hornblende separates gave minimum and maximum ages for intrusion of 108 ± 4 and 116 ± 4 Ma, respectively. A multi-whole-rock Rb–Sr isochron gave an age of 116 ± 20 Ma (Morrison et al. 1979). The neighbouring Mount McIntyre Pluton to the west intruded the Whitehorse Pluton (Fig. 3) and gives a U–Pb zircon date of $108.6^{+1.2}_{-0.4}$ Ma (Hart 1995), further limiting the minimum age. Thus the best estimate for the age of the Whitehorse Pluton is taken as 112 ± 4 Ma.

Geobarometry

The mean characteristic remanent magnetization (ChRM) vector is measured for a pluton in situ; however, if the pluton has suffered postmagnetization tilting, then the paleomagnetic estimates for latitudinal displacement and (or) rotation will be incorrect unless the vector is tilt corrected (Butler 1992; Ague and Brandon 1992). Amphibole geobarometry is employed here to estimate the amount and attitude of postmagnetization tilting of the Whitehorse Pluton.

Wones (1981) and Hammarstrom and Zen (1986) showed that the aluminum content of amphiboles in calc-alkalic granitic rocks is correlated with pressure of crystallization. The Al-in-hornblende geobarometer has been recalibrated for varying pressure and temperature ranges (Hollister et al. 1987; Johnson and Rutherford 1989; Blundy and Holland 1990; Schmidt 1992). Recently, Anderson and Smith (1995) discussed the sensitivity of this geobarometer to temperature and oxygen fugacity. Their revised calibration, which is based on the amphibole geobarometer of Schmidt (1992) and the hornblende–plagioclase geothermometer of Blundy and Holland (1990), is given as

$$P = 4.76\mathrm{Al}^{\mathrm{TOT}} - 3.01 - \frac{T - 675}{85}[0.530\mathrm{Al}^{\mathrm{TOT}} + 0.005294(T - 675)]$$

P is in MPa, error ±60 MPa; T is in °C.

To minimize the effects of the variation in oxygen fugacity, Anderson and Smith (1995) suggest that the $Fe^{3+}/(Fe^{3+} + Fe^{2+})$ ratio in the amphiboles should be greater than 0.20 and that the $Fe^{TOT}/(Fe^{TOT} + Mg)$ ratio should range between 0.40 and 0.65. The method outlined by Anderson and Smith (1995) requires an initial pressure to be calculated from the total aluminum in the hornblende, based on Schmidt's (1992) calibration:

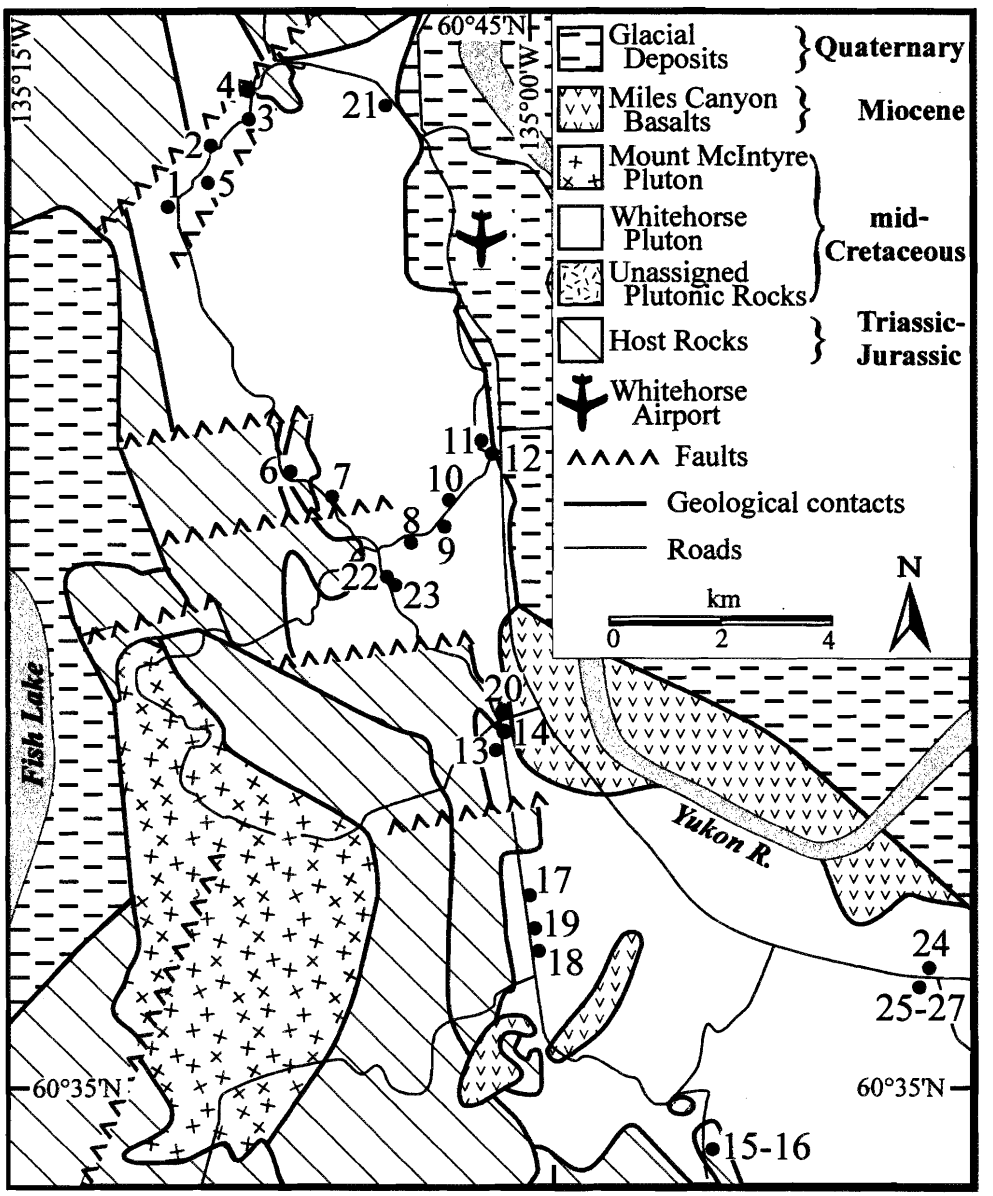
$$P = -3.01 + 4.76\mathrm{Al}^{\mathrm{TOT}} \qquad r^2 = 0.99$$

This pressure is then used in the Blundy and Holland (1990) geothermometer to obtain the temperature:

$$T = \frac{0.677P - 48.98 + Y}{-0.0429 - 0.008314 \ln K}$$

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Fig. 3. Geology of the Whitehorse Pluton after Hart and Pelletier (1989), with sampling sites WP01 (1) to WP27 (27).



where $K = [(Si - 4)/(8 - Si)]X_{ab}^{plag}$, and Y is the plagioclase nonideality function. This temperature and the total aluminum of the hornblende are then inserted into the Anderson and Smith (1995) calibration to obtain the temperature-corrected pressure.

Procedures

Nine polished thin sections of rocks from the north, central, and southern border zones of the Whitehorse Pluton (Tables 1, 2) were prepared for microprobe analysis. All nine thin sections carried the required mineral assemblage of plagioclase, hornblende, quartz, potassium feldspar, biotite, magnetite–ilmenite, titanite, and apatite. Chemical analyses on hornblende and plagioclase pairs were performed on the JEOL JXA-8600 Superprobe at the University of Western Ontario. Operating conditions were set at a 15 kV accelerating voltage, a 10 nA beam current, and a spectrum accumulation time of 20 s. Beam focus was 2 μm for amphibole and 5 μm for plagioclase. A minimum of two amphibole–plagioclase pairs were analyzed on each section, with a minimum of three analyses per crystal. Al-poor borders of amphibole grains were encountered within 20–30 μm of the grain edges. Similarly, albitic rims were found on plagioclase grains. These anomalous rim materials, presumably of subsolidus origin, could result in erroneous pressure estimates (Ague and Brandon 1996) and were avoided by traversing inward from the grain boundary until analyses gave a consistent set of compositions. Final compositions are the average of at least three spot analyses taken within 10 μm of each other.

Results

Representative compositions of amphibole grain edges used in the thermobarometric calculations are given in Table 1. Amphibole structural formulae are calculated on the basis

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Table 1. Representative amphibole analyses from the border zone of the Whitehorse Pluton.

Sample No.:	WP041	WP273	WP246	WP028	WP1753	WP0724	WP2136	WP1948	WP1212
Site No.:	WP04	WP27	WP24	WP02	WP17	WP07	WP21	WP19	WP12
SiO_2 (wt.%)	47.04	50.19	49.39	49.37	48.59	48.89	49.51	48.39	49.35
TiO_2	1.28	0.94	0.91	0.90	1.52	1.00	1.08	1.47	0.90
Al_2O_3	7.24	5.24	5.78	5.87	6.46	6.28	5.78	6.31	5.60
Fe_2O_3	3.03	6.07	6.52	4.44	5.70	4.98	4.90	6.54	6.55
Cr_2O_3	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.09	0.01
FeO	12.58	6.44	6.11	9.06	6.32	8.40	7.88	5.66	6.21
MnO	0.36	0.58	0.52	0.44	0.50	0.45	0.45	0.62	0.55
MgO	12.60	15.84	15.71	14.81	15.49	14.70	15.17	15.48	15.56
CaO	11.84	11.80	11.77	12.25	11.46	12.03	11.95	11.33	11.79
Na_2O	1.61	1.01	1.12	0.94	1.45	0.91	0.89	1.37	0.88
K_2O	0.57	0.32	0.29	0.42	0.30	0.51	0.47	0.36	0.41
H_2O	1.80	2.05	2.03	2.02	2.04	2.02	2.00	2.01	2.04
F	0.44	0.09	0.10	0.09	0.07	0.08	0.14	0.12	0.07
Cl	0.14	0.04	0.05	0.10	0.07	0.08	0.08	0.08	0.05
F, Cl = O	0.22	0.05	0.05	0.06	0.05	0.05	0.08	0.07	0.04
Total	100.34	100.56	100.25	100.64	99.92	100.29	100.22	99.76	99.92
Si (ppm)	6.897	7.156	7.070	7.101	6.981	7.049	7.113	6.965	7.093
$^{[4]}Al$	1.103	0.844	0.930	0.899	1.019	0.951	0.887	1.035	0.907
$^{[6]}Al$	0.148	0.037	0.045	0.096	0.076	0.116	0.092	0.036	0.042
Fe^{3+}	0.334	0.651	0.703	0.480	0.616	0.541	0.530	0.709	0.708
Fe^{2+}	1.543	0.768	0.731	1.089	0.759	1.013	0.947	0.681	0.746
Ti	0.141	0.101	0.098	0.097	0.164	0.108	0.117	0.159	0.097
Mg	2.753	3.366	3.351	3.174	3.317	3.159	3.248	3.321	3.333
Mn	0.045	0.070	0.063	0.054	0.061	0.055	0.055	0.076	0.067
Cr	0.002	0.000	0.000	0.000	0.000	0.000	0.000	0.010	0.001
Ca	1.860	1.803	1.805	1.888	1.764	1.858	1.840	1.747	1.816
$^{[M4]}Na$	0.140	0.197	0.195	0.112	0.236	0.142	0.160	0.253	0.184
$^{[A]}Na$	0.318	0.082	0.116	0.150	0.168	0.113	0.088	0.130	0.061
K	0.107	0.058	0.053	0.077	0.055	0.094	0.086	0.066	0.075
OH	1.761	1.950	1.943	1.935	1.951	1.944	1.917	1.926	1.956
F	0.204	0.041	0.045	0.041	0.032	0.036	0.064	0.055	0.032
Cl	0.035	0.010	0.012	0.024	0.017	0.020	0.019	0.020	0.012
Fe^{3+}	0.18	0.46	0.49	0.31	0.45	0.35	0.36	0.51	0.49
Fe	0.36	0.19	0.18	0.26	0.19	0.24	0.23	0.17	0.18

Notes: Structural formulae are normalized to 13 cations exclusive of Ca, Na, and K. Fe_2O_3, H_2O, and OH are back-calculated from the structural formula. $Fe^{3+} = Fe^{3+}/(Fe^{2+} + Fe^{3+})$; $Fe = Fe^{2+}/(Fe^{2+} + Mg)$.

of 13 cations. Cosca et al. (1991) show that the 13-cation normalization closely predicts site occupancies and gives reasonable estimates of Fe^{3+} and Fe^{2+}. As Anderson and Smith (1995) point out, the choice of a particular amphibole normalization scheme does not affect Al^{TOT} and thus the pressure using the Schmidt (1992) calibration. It can, however, affect the $^{[4]}Al$ and $^{[6]}Al$ distribution, and thus the thermometric calculations and the final pressures calculated using the Anderson and Smith (1995) calibration.

The amphiboles exhibit a small compositional range (Table 1; Fig. 4*a*) and all are classified as magnesiohornblende. Only two crystals, both from site WP04, are somewhat less siliceous and more aluminous than the main compositional group. $Fe^{3+}/(Fe^{3+} + Fe^{2+})$ ratios range from 0.3 to 0.5, except at site WP04, where the ratio falls to 0.2 (Table 1). Total iron/magnesium ratios in the amphiboles are generally about 0.3. The amphiboles at site WP04 are, however, significantly less magnesian (Table 1; Fig. 4*a*).

The temperature-sensitive edenite exchange reaction albite + tremolite = 4 quartz + edenite, i.e., $Si + {}^{[A]}\square = {}^{[4]}Al + {}^{[A]}(Na + K)$, is tested in Fig. 4*b*, where $^{[A]}\square$ stands for an A-site vacancy. The pressure-sensitive tschermakite-forming reaction 2 quartz + 2 anorthite + biotite = orthoclase + tschermakite ($^{[4]}Al + {}^{[6]}Al = Si + R^{2+}$) is tested in Fig. 4*c*. Clearly, both reactions are important in the border zones of the Whitehorse Pluton. Using the calibration of Schmidt (1992), the amphiboles indicate pressures ranging from 110 to 310 MPa (Fig. 4*c*), with a concentration of data near 200 MPa. The geobarometer is experimentally calibrated between 200 and 1300 MPa (Anderson and Smith 1995). The reason for the lower end is calc-alkalic magmas become temperature sensitive with pressure changes below 200 MPa (Hollister et al. 1987). However, in this study, it is the gradient in pressures across the pluton that we are concerned about and, therefore, only the relative precision of the pressure values is required, not the absolute accuracy.

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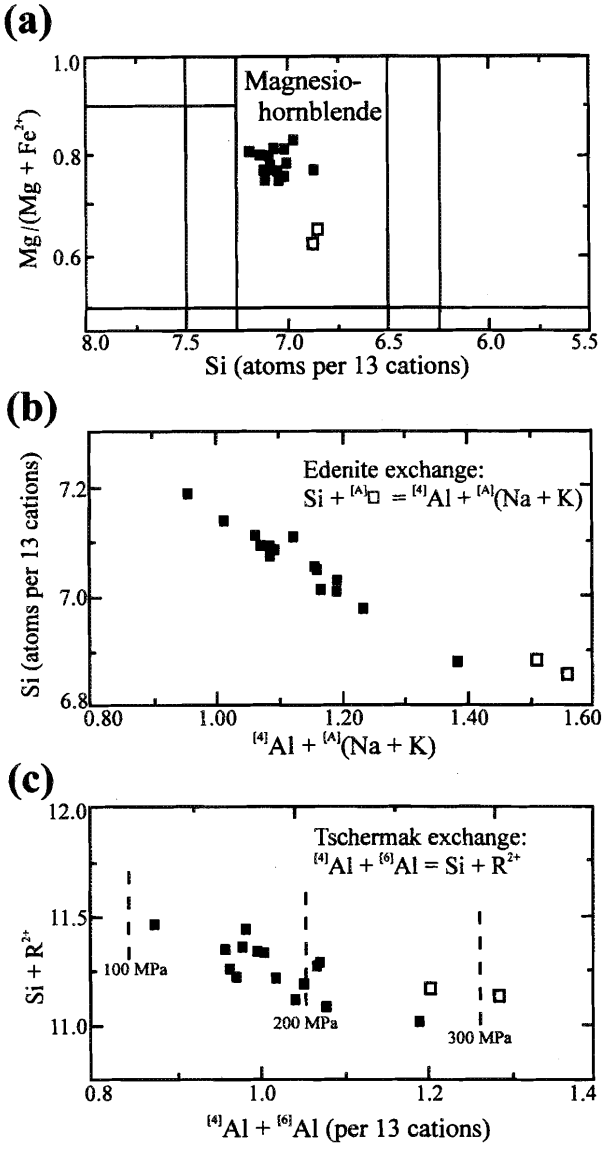


Fig. 4. Hornblende compositions of the Whitehorse Pluton. □, site WP04. (*a*) Simplified calcic-amphibole classification plot (after Hawthorne 1981). (*b*, *c*) Amphibole exchange reactions. Pressures shown in (*c*) are based on the Schmidt (1992) calibration.

Site averages of amphibole–plagioclase pairs for each collection site in the Whitehorse Pluton border zones were calculated and the final thermobarometric calculations were made using these data (Table 2). Final pressures were estimated using the expression given by Anderson and Smith (1995) and intrusion depths were estimated from these pressure data and an assumed crustal density of 2.8 g/cm^3. Site-averaged temperatures range from 665 to 714°C (Table 2), and temperature-corrected pressures range from 155 to 276 MPa (Fig. 5; Table 2). It should be noted that the 276 MPa pressure obtained for site WP04 is anomalous. Geobarometric measurements for all other border-zone samples indicate a very small range of pressures around 180 ± 20 MPa (Fig. 5).

Paleomagnetism

Methods

Twenty-seven sites were sampled across the pluton to represent it both petrographically and spatially (Fig. 3). At least five cores per site were drilled and oriented in situ, generally using a solar compass. Sites WP06, WP15, and WP16 provided paleomagnetic contact tests between the pluton and its clastic and carbonate host rocks of the Lewes River Group. Site WP25 is a mafic dike and site WP26 is in its granitoid baked zone. The other 22 sites are from the granitoid rocks of the Whitehorse Pluton.

All paleomagnetic measurements were done at the University of Windsor using an automated CTF DRM-420 cryogenic magnetometer, a Sapphire Instruments SI-4 alternating-field (AF) demagnetizer, a Sapphire Instruments SI-5 spinner magnetometer, a Sapphire Instruments SI-6 pulse magnetizer, and a Magnetic Measurements MMTD1 thermal demagnetizer. All the instruments are in a magnetically shielded room that has an ambient field of 0.2% of the Earth's magnetic field intensity.

The natural remanent magnetization (NRM) was measured for each of the 290 specimens. Two specimens with average NRM intensities and directions were selected as pilot specimens for each site. One pilot specimen was subjected to AF demagnetization in 10 steps from 5 to 130 mT and the other to thermal demagnetization in 11 steps from 200 to 600°C. The results for the pilot specimens showed that all but one site had magnetic coercivity and unblocking temperature spectra that are typical of magnetite (Fig. 6). Site WP14 showed thermal unblocking temperature spectra that are indicative of hematite. Thus, for time efficiency, all remaining specimens except for those from site WP14 were subjected to AF demagnetization, using at least five steps from 20 or 30 mT up to 60, 80, 100, or 120 mT (Table 3). The specimens of site WP14 were AF demagnetized in five steps up to 100 mT and then thermally demagnetized in three steps from 600 to 660°C. Saturation isothermal remanent magnetization (SIRM) testing was done on 14 specimens representing the range of rock types. Each specimen was pulse magnetized in 11 steps up to 900 mT and then AF demagnetized in seven steps to 140 mT. Specimen ChRM directions were obtained using the method of Kirschvink (1980) and were accepted for statistical analysis if the best-fit stable direction had a maximum angular deviation of $<15°$ over three or more steps, with most giving far smaller deviations. Site and unit mean directions were calculated using the statistical method of Fisher (1953) (Table 3).

Results

Pluton mean direction

No significant differences were noted in the magnetic characteristics of the five rock types sampled in the Whitehorse Pluton. A minor pyrrhotite component can be seen on thermal demagnetization in many of them (Fig. 6). The SIRM acquisition and decay curves for all specimens show pseudo-single-domain to multidomain magnetite characteristics (Fig. 7). The ChRM vectors of all specimens show normal polarities, as they should for ~112 Ma rocks of the Cretaceous normal-polarity superchron (Gradstein et al. 1994). A

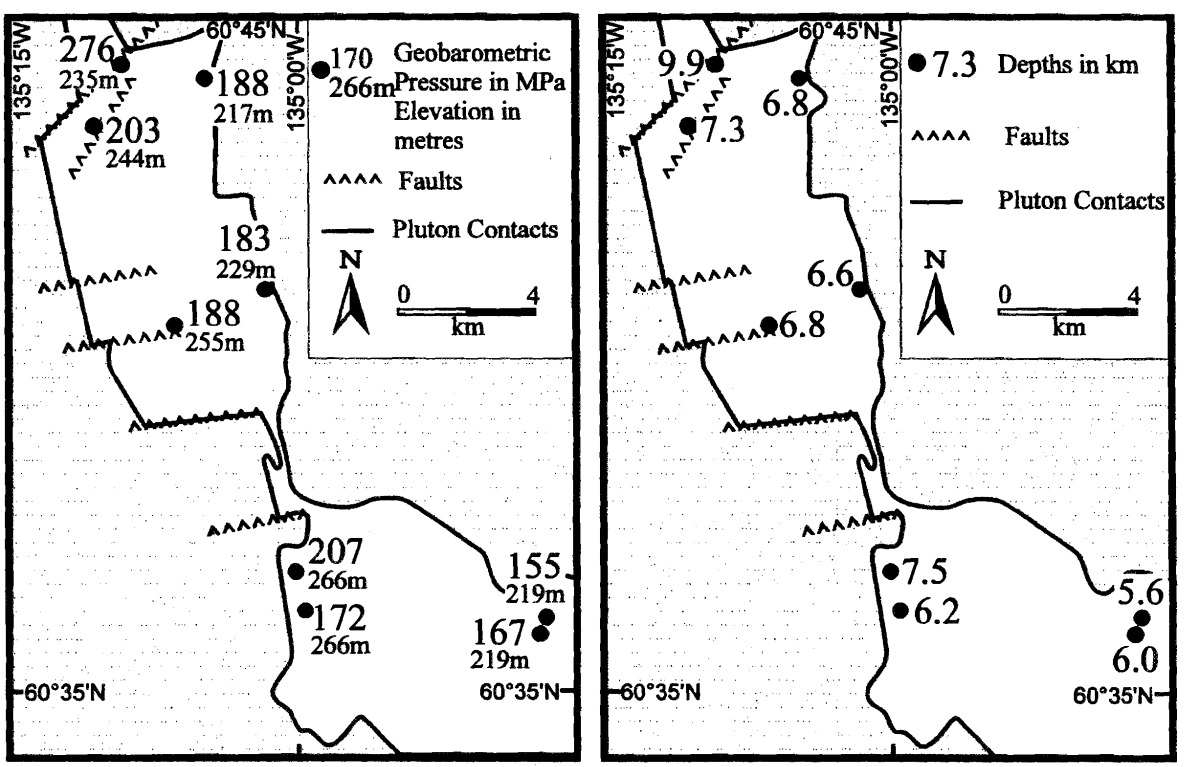
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Table 2. Site-averaged compositions of coexisting amphibole and plagioclase edges and thermobarometric data for the Whitehorse Pluton.

Site	Plagioclase Ab #[a]	Amphibole Si	Amphibole Al^{TOT}	P (MPa) Schmidt (1992)	P (MPa) Anderson and Smith (1995)	T (°C)[b]	Depth (km)
WP02	0.71	10.89	1.067	210±70	203	680	7.3
WP04	0.81	11.28	1.238	290±50	276	689	9.9
WP07	0.77	11.13	1.017	180±30	188	667	6.8
WP12	0.79	11.23	1.004	180±40	183	665	6.6
WP17	0.72	10.83	1.145	240±30	207	714	7.5
WP19	0.73	10.95	1.007	180±40	172	684	6.2
WP21	0.75	10.97	1.024	190±20	188	672	6.8
WP24	0.71	10.87	0.971	160±10	155	684	5.6
WP27	0.74	10.96	0.980	170±70	167	673	6.0

[a]Atomic ratio [Na/(Na + Ca)] in plagioclase.
[b]From Blundy and Holland (1990).

Fig. 5. Geobarometry, topographic elevation, and intrusion-depth data. Site-averaged pressures are from the Anderson and Smith (1995) calibration (Table 3).



steeply downward and northwards ChRM direction was isolated in nearly all specimens from 22 of the 24 granitoid sites after removal of a viscous component by ~20 mT or ~200°C (Fig. 8). The unit mean ChRM for the pluton gives a declination (D) of 16.7°, inclination (I) of 76.2°, radius of 95% confidence (α_{95}) of 3.1°, and precision parameter (k) of 101 (Fig. 9; Table 3). To be certain that any biasing effect of viscous remanent magnetization (VRM) was removed, the analysis was rerun using specimen vectors defined by coercivities over ~50 mT and unblocking temperatures over ~500°C. All specimens but two that were used in the initial analysis gave acceptable vectors, and their median initial intensity was still 29% of their median NRM intensity of 0.39 A/m. These vectors give a unit mean direction of D = 18.3°, I = 76.7°, α_{95} = 3.1°, and k = 81 which is a statistically insignificant 0.5° from the initially accepted unit mean ChRM direction, and shows that the VRM had been thoroughly erased.

Site WP14 gives a different direction from the pluton, with a mean of D = 16.7°, I = 9.6°, α_{95} = 5.3°, and k = 67 (Fig. 9; Table 3). As described earlier, these specimens were AF then thermally demagnetized to isolate their

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Table 3. Site-averaged paleomagnetic data for the Whitehorse Pluton.

Site[a]	N	n	Demagnetization steps: AF (mT)	Demagnetization steps: Thermal (°C)	D (°)	I (°)	α_{95}	k
WP01	8	8	20–120	200–500	349.6	74.9	3.2	310
WP02	8	8	20–100	200–600	330.7	76.7	3.6	242
WP03	8	7	20–80	200–320	359.2	76.4	3.7	261
WP04	9	9	20–130	200–600	328.2	80.2	4.8	117
WP05	9	9	20–100	200–600	6.5	77.4	2.5	418
WP06d	6	6	20–100	200–600	74.2	80.0	6.0	124
WP06h	11	10	20–100	200–600	30.2	78.4	7.7	40
WP07	12	12	20–60	200–600	22.5	69.0	4.0	118
WP08	12	12	20–100	200–600	41.7	71.6	2.2	398
WP09	10	10	20–60	200–560	355.0	69.5	5.2	89
WP10	10	10	20–100	200–600	29.7	85.5	5.0	94
WP11	10	10	20–100	200–600	12.6	74.4	3.7	170
WP12	12	12	20–80	200–600	39.4	79.1	2.9	223
WP13	14	14	20–80	260–575	32.1	71.9	2.9	120
WP14ag	14	12	20–100	600–660	16.7	9.6	5.3	67
WP15rb	14	13	20–100	200–540	53.6	75.9	8.6	24
WP16u	15	6	20–60	260–400	337.0	43.4	12.2	31
WP16c	15	6	20–60	260–400	15.4	17.4	12.2	31
WP17	5	5	20–60	200–560	15.2	76.1	6.0	163
WP18	12	12	20–100	200–600	358.5	68.3	2.9	231
WP19	9	9	30–60	200–575	15.9	73.9	4.6	125
WP20	10	8	20–60	200–600	22.7	59.8	5.0	122
WP21	10	10	20–100	200–600	341.7	74.1	1.9	634
WP22	9	8	30–60	200–600	51.3	76.2	4.4	163
WP23	6	6	20–60	200–600	29.2	75.6	5.2	168
WP24	10	10	30–60	200–600	351.1	77.9	4.4	124
WP25md	14	9	20–100	400–560	14.9	73.1	11.7	20
WP26ag	10	8	20–100	200–565	117.1	21.5	9.8	33
WP27	9	7	20–100	200–600	31.4	80.7	7.8	60
Pluton[b]	24	22			16.7	76.2	3.1	101

Notes: The mean direction is given by its declination (*D*), inclination (*I*), radius of 95% confidence (α_{95}), and precision parameter (*k*) (Fisher 1953). *N*, number of specimens measured; *n*, number of specimens used to compute the averages.

[a]ag, altered granite; c, tilt corrected; d, granite dike; h, marble host rock; md, mafic dike; rb, red bed; u, uncorrected mean.

[b]Excludes means for sites WP06h, WP14ag, WP15rb, WP16u, WP16c, WP25md, and WP26ag.

hematite component. Petrographically, they are recrystallized, leucocratic, and contain more carbonate and plagioclase than all the other sites. Hence, these rocks were deemed to be highly altered, and their site mean ChRM direction was not included in the overall ChRM mean of the pluton.

Contact tests

Sites WP06, WP15, and WP16 were collected as a paleomagnetic contact test. At site WP06, a 40 cm wide, fine-grained granodiorite dike intrudes marble of the Lewes River Group within a few metres of its contact with the Whitehorse Pluton. The dike (site WP06d) and marble (site WP06h) gave ChRM directions similar to that of the main pluton (Fig. 9; Table 3). Site WP15 is in a hematitic metasiltstone and site WP16 is in a dolostone, with both being about 100 m from the Whitehorse Pluton (Fig. 3). The hematite alteration in the metasiltstone gives the same ChRM direction as the pluton before tectonic correction; however, the magnetite-bearing dolostone gives a different direction both before and after tilt correction. The latter direction is shallow to the northeast (Fig. 9; Table 3) and may define the direction of the last magnetization event for the Late Triassic Lewes River Group. Taken together these sites are thought to record a positive paleomagnetic contact test that indicates that the Whitehorse Pluton retains a primary remanence.

Site WP25 is located in one of the fine-grained vertical diabase dikes of unknown age that cut the pluton. It was collected in the hope that it might be a reversed-polarity Miocene dike. Unfortunately, it gives the same normal-polarity ChRM direction as the adjacent host rock at site WP27 and as the pluton overall (Fig. 9; Table 3). Thus, the dike could be a late phase of the pluton or of Miocene or some other age. Therefore, this contact test is inconclusive.

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Fig. 6. Thermal demagnetization decay curves for the Whitehorse Pluton. Numbers refer to the site of the treated specimen. The preferred unblocking temperature diagnostic of pyrrhotite and magnetite are indicated by P and M, respectively. J/J_{NRM} is the ratio of the measured remanence intensity to NRM intensity.

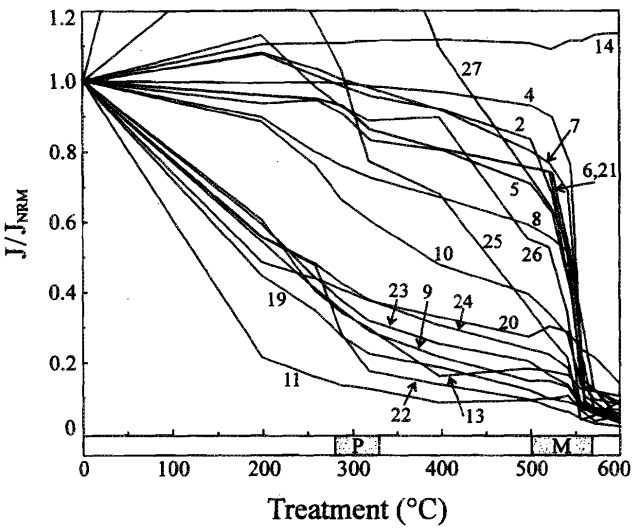
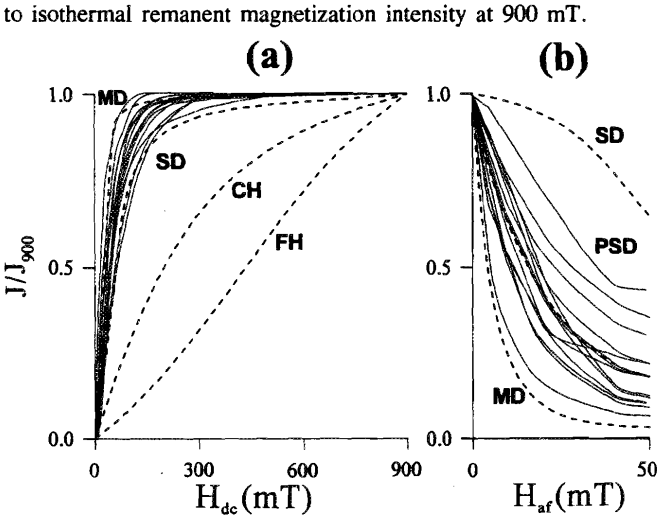


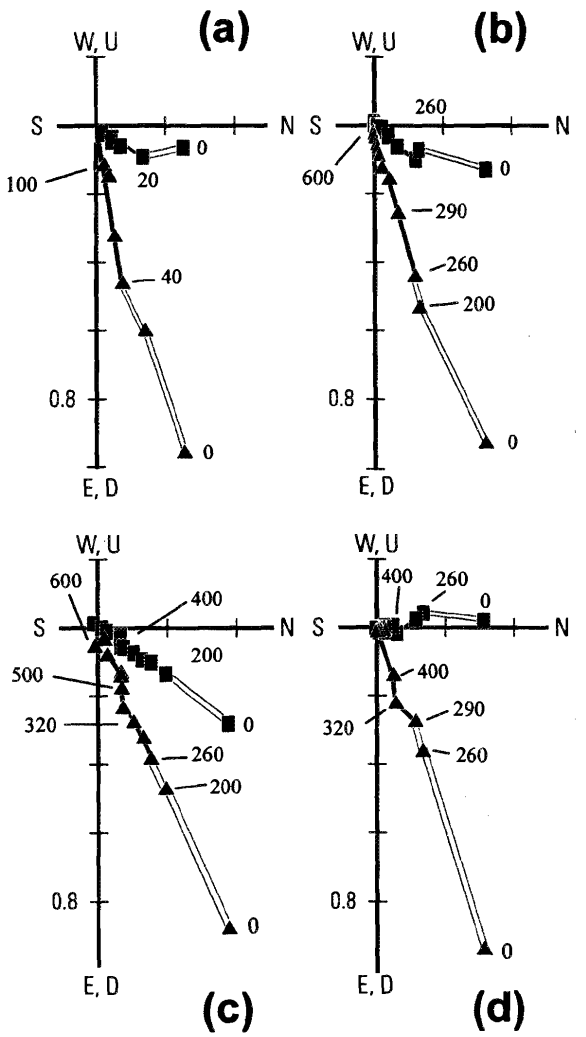
Fig. 7. SIRM acquisition (*a*) and decay (*b*) curves for the Whitehorse Pluton. H_{dc} and H_{af} are the applied magnetizing and demagnetizing fields, respectively. The type curves are for single-domain (SD), pseudo-single-domain (PSD), and multidomain (MD) magnetite and fine-grained (FH) and coarse-grained (CH) hematite. J/J_{900} is the ratio of the measured remanence intensity to isothermal remanent magnetization intensity at 900 mT.



Discussion and conclusions

The Whitehorse Pluton is thought to be posttectonic (Hart 1995). Igneous textures and mineralogy indicate that the pluton has not been metamorphosed or heated to high enough temperatures to affect the hornblende chemistry (>600°C), the magnetic properties (>450°C), or the K–Ar systematics in biotite (>280°C). Local weak foliations are magmatic in origin. Hydrothermal alteration is restricted to the margins of the pluton and caused by interactions with meteoric water (Morrison 1981). Therefore, the measured pressures, temperatures, and ChRM directions are considered primary.

Fig. 8. Representative orthogonal step demagnetization plots. (*a*) Specimen is AF demagnetized (measured in mT). (*b*–*d*) Specimens are thermally demagnetized (measured in °C). ■, projected in the north–east–south–west plane; ▲, projected in the up–north–down–south plane. Double lines denote the VRM component, and thick lines denote the ChRM component.

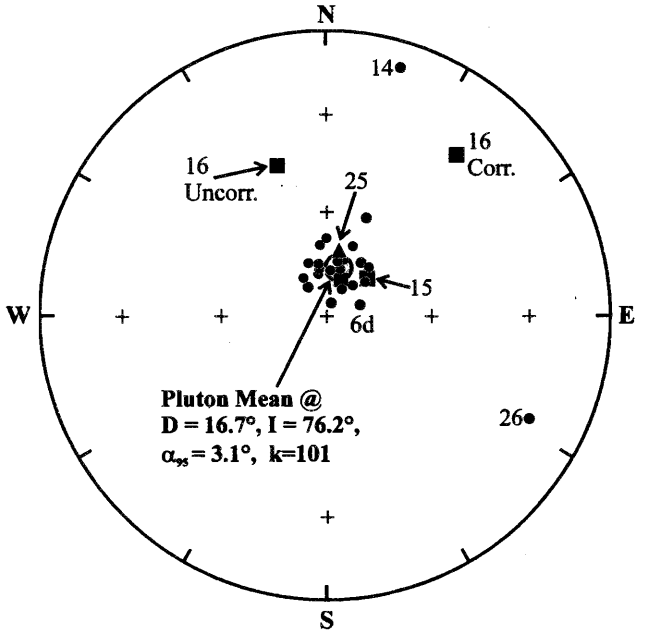


Geobarometry

The maximum topographic relief between the eight geobarometry sites in the Whitehorse Pluton is only 49 m (Fig. 5), yielding a topographic correction of less than 1.5 MPa. Except for an anomalous result at site WP04, the geobarometric measurements indicate a small range of pressures corresponding to shallow depths of intrusion from 5.6 to 7.5 km, with a mean of 6.6 ± 0.6 km (Fig. 5; Table 2). The presence of roof pendants along the pluton's margins is also an indication of shallow depth. Noting that the error in any individual determination is ±60 MPa (Anderson and Smith 1995), the expected accuracy of this method is equivalent to ±2.1 km, which is far greater than the determined value of ±0.6 km. Thus, there is no reason to believe that the pluton has been significantly tilted. If the accepted values are averaged into northerly (sites WP02 and WP21), central (sites WP07 and WP12), and southerly sections (sites WP17, WP19, WP24, and WP27), they give means of 7.1, 6.7, and

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Fig. 9. Equal-area stereonet of all site means of the Whitehorse Pluton (Table 4). ●, granitoids; ■, host rocks; ▲, mafic dike. The mean is a dot in the middle of the shaded circle, which defines the cone of 95% confidence of Fisher (1953).



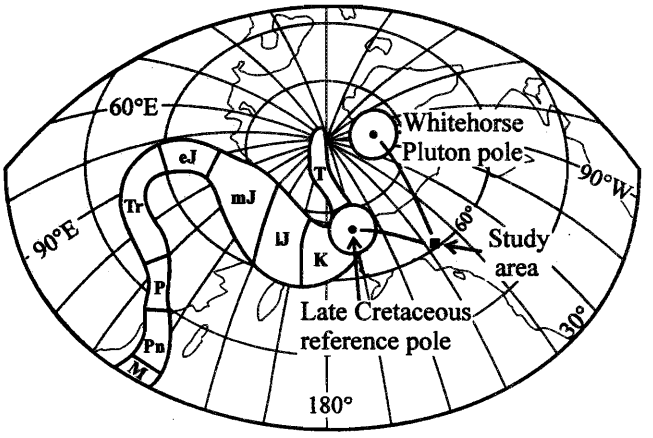
6.3 km, respectively, indicating an insignificant south-south-easterly down tilt of ~0.8 km, which is equivalent to ~2.3° of arc. This is less than the 3.1° cone of 95% confidence around the pluton's mean ChRM direction, but, if a tilt correction were made, then the mean inclination would steepen and decrease the northward translation estimate by ~250 km. The rocks at site WP04 are highly altered, oxidized, and deeply weathered, and the site is adjacent to a fault and near a copper skarn deposit. Thus, it is probable that wall-rock exchange reactions have produced the anomalous amphibole and plagioclase compositions at this site.

Tectonics

The ~112 Ma Whitehorse Pluton's mean ChRM direction corresponds to a paleopole of 285.5°E, 81.7°N (d_p = 5.3, d_m = 5.7). This pole is compared with the 112 Ma reference pole for the North American craton of 194.6°E, 69.5°N (α_{95} = 5.1) (Fig. 10), which is interpolated from the apparent polar wander path (APWP) of Besse and Courtillot (1991). The comparison estimates that the pluton has been translated polewards or generally northwards by 11.0 ± 4.8°, equivalent to 1220 ± 530 km, and rotated clockwise by 59 ± 17° (Table 4). Well-determined pole positions are available for four other Cretaceous to latest Jurassic units in the terranes underlying the Intermontane Belt, i.e., the Carmacks volcanics, Axelgold intrusions, Spences Bridge volcanics, and Francois Lake (Topley) intrusions (Fig. 1). Their poles have also been compared to their corresponding reference pole on the North American APWP of Besse and Courtillot (1991) (Table 4).

All five collections indicate northward translations of between 1220 and 2240 km; however, such a direct comparison may be misleading because it does not incorporate time.

Fig. 10. Locations of the Whitehorse Pluton study area, unit mean pole, Late Cretaceous reference pole from Besse and Courtillot (1991), and APWP of the North American craton from Van der Voo (1993).



A more enlightening method is to compare the average rates of northward motion. First, it is assumed that northward motion relative to the craton ended with accretion in Eocene time at about 45 Ma (Symons and Welling 1989; Bardoux and Irving 1989; Vandall and Palmer 1990; Irving et al. 1996). Then the rate is calculated by dividing the translation distance by the duration of time between 45 Ma and the age of the unit (Table 4). Excluding the result for the Carmacks Group volcanics, which will be discussed separately, the remaining four units give remarkably consistent rates between 1.8 ± 0.8 and 2.7 ± 1.3 cm/a. These values are not significantly different at the 95% confidence level and, therefore, can be validly combined to give a minimum average rate of 2.3 ± 0.4 cm/a for the Intermontane terranes (Fig. 11). This is a typical rate for plate motion on the Earth's surface. This is a minimum rate because it does not include the paleolongitudinal component of motion and an average rate because it does not compensate for probable changes in the rate of translation from ~143 to ~45 Ma (Engebretson et al. 1985). Given that the four rock units span the Intermontane Belt from southern British Columbia to southern Yukon, the data indicate that the terranes underlying the Intermontane Belt behaved as a relatively coherent unit during their northward trek along the North American margin. Thus, they do not support the model of Irving et al. (1996), who suggested that the terranes underlying the northern portion of the Intermontane Belt travelled ~2000 km northward on dextral transcurrent faults, whereas those in the southern part travelled only ~1000 km northward.

The estimates for rotation derived from the paleomagnetic poles show more scatter. The Axelgold intrusions, Spences Bridge volcanics, and Whitehorse Pluton are statistically the same, indicating about 60 ± 15° of clockwise rotation. Again, these values span the length of the Intermontane Belt, implying that the Intermontane terranes probably behaved as a fairly coherent unit during their travels. Both the Francois Lake intrusions and the Carmacks Group volcanics showed only minor amounts of rotation (Table 4). The cause of the former's discordance is not yet clear; however, the latter

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Fig. 11. Plot of paleolatitude for the Intermontane Belt vs. time. Error bars on the blocks represent the 95% confidence errors (Table 4).

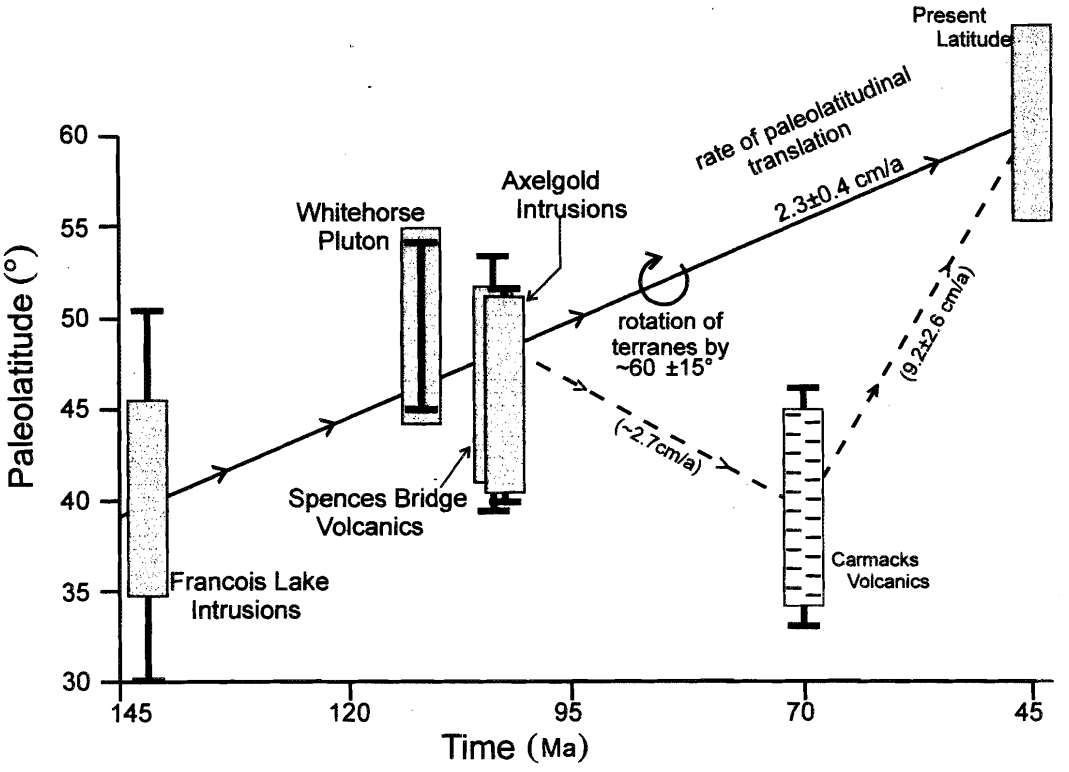


Table 4. Displacements of Cretaceous rock units of the Intermontane Superterrane.

Unit	Location	Age (Ma)	Paleolatitude (°N)[a]	Translation (km)[a]	Average rate of motion (cm/a)[b]	Rotation (°)[a,c]	Reference
Carmacks volcanics	South Yukon	70	20.6±7.4	2290±650	9.2±2.6	20±10 CW	Johnston et al. 1996
Axelgold intrusions	Central British Columbia	103	13.9±5.9	1540±650	2.7±1.3	55±13 CW	Monger and Irving 1980
Spences Bridge volcanics	South British Columbia	104	13.5±7.4	1500±820	2.5±1.3	66±13 CW	Irving and Thorkelson 1990; Irving et al. 1995
Whitehorse Pluton	South Yukon	112	11.0±4.8	1220±530	1.8±0.8	59±17 CW	This study
Francois Lake intrusions	Central British Columbia	142	20.2±10.8	2240±1200	2.3±1.2	3±17 CCW	Symons 1983

[a]Calculations are from Butler (1992, Appendix A), with reference poles interpolated from Besse and Courtillot (1991).
[b]Rates are calculated by dividing the translation distance by the age less the accretion age (45 Ma) (Irving et al. 1996).
[c]CW, clockwise; CCW, counterclockwise.

could postdate the clockwise rotation event (Fig. 11).

The Carmacks Group volcanics has given a high but tenable rate of paleolatitude displacement of 9.2 ± 2.6 cm/a (Table 4) (Marquis and Globerman 1988; Johnston et al. 1996), which is statistically significantly different from the other four values. Also, this unit has given a rotation estimate that is significantly different from those of three of the other four units. These differences may reflect the failure of the Carmacks volcanics to record the time-averaged Earth's magnetic field direction with fidelity for some reason. The significant difference in the overall precision parameter for reversely polarized sites compared with normally polarized sites and the significant directional differences between different sections of flows hint at possible problems. Another possibility is that the motion of the Intermontane terranes relative to the craton was more complicated, moving relatively slowly northward in the Late Jurassic – Early Cretaceous (Fig. 11), then southward during the Late Cretaceous, and then very rapidly northward during Paleocene–Eocene time (Engebretson et al. 1995). A paleomagnetic study is now in progress by the authors on a latest Cretaceous pluton to check the Carmacks' result.

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Conclusions

This paleomagnetic and geobarometric study of the Whitehorse Pluton estimates that the Stikinia terrane in the Yukon was translated northwards by 1220 ± 530 km and rotated clockwise by 59 ± 17°. Comparison with other Early Cretaceous results suggests that the terranes underlying the Intermontane Belt behaved as a relatively coherent unit, which moved northward at a minimum average rate of ~2.3 ± 0.4 cm/a for the interval from ~142 to ~45 Ma and rotated clockwise by ~60° at some time between ~100 and ~70 Ma during their northward trek. More paleomagnetic data are needed for the northern Cordillera to better constrain these estimates.

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